

Data types:

To store anything in our system, we should have allocate the memory [bytes]. Then having the properties.

1. What type of data we want to store in that memory.
2. How many bytes required.
3. How much value we want to store [range]

These 3 properties decided by the data type.

Primitive data type: single valued data types

`int sal = 30000;` ← non-decimal no

short int / signed int / int c & c++ int capacity → **-32768 to +32767** → 16 bit compilers → 2^{16} → $65536/2$ → + / - → 2 bytes → **%d** ← **conversion char / format specifier**

in java / .net / py → -2147483648 to +2147483647 → 2^{32} → 4294967296 → 4 bytes

unsigned int sal = 60000; → 0 to 65535 → 2 bytes → **%u**

signed long int sal = 250000; → **-2147483648 to +2147483647** → 4 bytes → **%ld**

unsigned long int → 0 to 4294967295 → 4 bytes → **%ul**

float height = 5.2; → **%f** → 4 bytes → 3.4×10^{-38} to $3.4 \times 10^{+38}$

long float / double → %lf → 8 → 1.7×10^{-308} to $1.7 \times 10^{+308}$

long double → %Lf → 10 → 3.4×10^{-4932} to $1.1 \times 10^{+4932}$

char data type → 1 byte → 256 characters ← ASCII values → 2^8
→ 1 byte

1. **Signed char** → -128 to +127 → %c

2. **Unsigned char** → 0 to 255 → %c

String ← non-primitive

`char name[ 10] = "kishore.p";` ← **String** ← group of char's

`char name[ ] = "kishore.p";` ← **String** ← group of char's

2. derived data types - They are created from primitive data types.

1. **array** ← `int a[10];` ← int array variable ← non-primitive

`Char gender[10]="Male";` ← char variable ← **String**

2. **pointer** ← `int * p;` ← int pointer variable

3. **function** → `int sum( );` ← int function

User defined data types:

1. Structure
2. Union
3. Enum / enumeration

DATA TYPES

Data type determines the type of value we are going to store in our computer. To store anything in our computer, we should have to allocate the memory. This memory allocation is depended on the data type we are using.

Data type determines the properties such as

1. No of bytes
2. Range
3. Type of value

In C language we are having 3 **basic** data types

- 1. Int – To store non-decimal numbers**
- 2. Float – To store decimal numbers**
- 3. Char – To stores alphabets, numbers and special char**

Total data types are divided into 3 types.

- 1. Primitive data types**
- 2. Derived data types**

3. User defined data types

PRIMITIVE DATA TYPES:

These are the regular data types we are using in our c programs.

Data type	Bytes	Conversion Character / format specifier	Storage Range
int / signed int / short int	2	%d	-32768 to +32767
unsigned int	2	%u	0 to 65535
long int	4	%ld	-2147483648 to 2147483647
unsigned long int	4	%lu	0 to 4294967295
float	4	%f	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
double	8	%lf	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
long double	10	%Lf	$3.4 * 10^{-4932}$ to $1.1 * 10^{+4932}$
char	1	%c	1 character Signed char [-128 to +127] Unsigned char [0 to 255]
char[10] (STRING)	10	%s	9 char + 1 null char
void [empty			nothing

data type]			
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DERIVED DATA TYPES:

They are derived from primitive data types.

- 1. Array [non-primitive]**
- 2. Pointer**
- 3. Function**

USER DEFINED DATA TYPES:

These are the data types created by the user.

1. structure
2. union
3. enum